

Substitution Rule - Continue

$$\textcircled{1} \int \frac{x}{x^2+1} dx = \int \frac{du/2}{u} = \frac{1}{2} \ln|u| + C = \frac{1}{2} \ln|x^2+1| + C \\ = \ln\sqrt{x^2+1} + C$$

$$x^2+1=u \quad 2x dx = du \quad x dx = \frac{du}{2}$$

$$\textcircled{2} \int \frac{\sin(3\ln x)}{x} dx = \int \sin u \cdot \frac{du}{3} = \frac{1}{3} (-\cos u) + C$$

$$\left. \begin{array}{l} \text{Let } u = 3\ln x \\ du = 3 \frac{dx}{x} \Rightarrow \frac{dx}{x} = \frac{du}{3} \end{array} \right\} = -\frac{1}{3} \cos(3\ln x) + C$$

$$\textcircled{3} \int e^x \sqrt{1+e^x} dx = \int v^{1/2} \cdot dv = \frac{2}{3} (1+e^x)^{3/2} + c$$

$$1+e^x = v \Rightarrow e^x dx = dv$$

$$\textcircled{4} \int \frac{dx}{x^2+4x+5} = \int \frac{dx}{(x+2)^2+1} = \int \frac{dt}{t^2+1} = \text{Arctan}(x+2) + c$$

$$x+2 = t$$

$$dx = dt$$

$$(5) \int \frac{dx}{\sqrt{e^{2x}-1}} = \int \frac{dx}{e^x \sqrt{1-e^{-2x}}} = \int \frac{e^{-x} dx}{\sqrt{1-(e^{-x})^2}}$$

$$e^{-x} = u$$

$$-e^{-x} dx = du$$

$$e^{-x} dx = -du$$

$$= \int \frac{-du}{\sqrt{1-u^2}} = -\arcsin u + C$$

$$= -\sin^{-1}(u) + C$$

$$= -\arcsin(e^{-x}) + C$$

$$= -\sin^{-1}(e^{-x}) + C$$

$$\text{Ex: } \int \frac{3\sqrt{1+\tan^{-1}x}}{2+2x^2} dx = \frac{3}{2} \int \sqrt{u} du.$$

$$1+\tan^{-1}x = u$$

$$\frac{1}{1+x^2} dx = du$$

$$= \frac{3}{2} \cdot \frac{2}{3} \cdot u^{3/2} + C$$

$$= (1+\tan^{-1}x)^{3/2} + C$$

$$\text{Ex: } \int \frac{e^{\arctan x}}{1+x^2} dx = ?$$

$$u = \arctan x$$

$$x = \tan u$$

$$1 = u' (1 + \tan^2 u)$$

$$\frac{du}{dx} = u' = \frac{1}{1+\tan^2 u} \Rightarrow du = \frac{dx}{1+x^2}$$

$$\int \frac{e^{\arctan x}}{1+x^2} dx = \int e^u + C = e^{\arctan x} + C$$

Every differentiation rule has a corresponding integration rule. For instance, the Substitution Rule for integration corresponds to the Chain Rule for differentiation. The rule that corresponds to the Product Rule for differentiation is called the rule for *integration by parts*.

The Product Rule states that if f and g are differentiable functions, then

$$\frac{d}{dx} [f(x)g(x)] = f(x)g'(x) + g(x)f'(x)$$

In the notation for indefinite integrals this equation becomes

$$\int [f(x)g'(x) + g(x)f'(x)] dx = f(x)g(x)$$

or

$$\int f(x)g'(x) dx + \int g(x)f'(x) dx = f(x)g(x)$$

We can rearrange this equation as

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$$\int f(x)g'(x) dx = f(x)g(x) - \int g(x)f'(x) dx$$

Formula 1 is called the **formula for integration by parts**. It is perhaps easier to remember in the following notation. Let $u = f(x)$ and $v = g(x)$. Then the differentials are $du = f'(x) dx$ and $dv = g'(x) dx$, so, by the Substitution Rule, the formula for integration by parts becomes

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$$\int u dv = uv - \int v du$$

EXAMPLE 1 Find $\int x \sin x \, dx$.

SOLUTION USING FORMULA 1 Suppose we choose $f(x) = x$ and $g'(x) = \sin x$. Then $f'(x) = 1$ and $g(x) = -\cos x$. (For g we can choose *any* antiderivative of g' .) Thus, using Formula 1, we have

$$\begin{aligned}\int x \sin x \, dx &= f(x)g(x) - \int g(x)f'(x) \, dx \\&= x(-\cos x) - \int (-\cos x) \, dx \\&= -x \cos x + \int \cos x \, dx \\&= -x \cos x + \sin x + C\end{aligned}$$

It's wise to check the answer by differentiating it. If we do so, we get $x \sin x$, as expected.

SOLUTION USING FORMULA 2 Let

$$u = x \qquad dv = \sin x \, dx$$

$$\text{Then} \qquad du = dx \qquad v = -\cos x$$

and so

$$\begin{aligned}\int x \sin x \, dx &= \int \overbrace{x}^u \overbrace{\sin x \, dx}^{dv} = \overbrace{x}^u \overbrace{(-\cos x)}^v - \int \overbrace{(-\cos x)}^v \overbrace{dx}^{du} \\&= -x \cos x + \int \cos x \, dx \\&= -x \cos x + \sin x + C\end{aligned}$$

NOTE • Our aim in using integration by parts is to obtain a simpler integral than the one we started with. Thus, in Example 1 we started with $\int x \sin x \, dx$ and expressed it in terms of the simpler integral $\int \cos x \, dx$. If we had chosen $u = \sin x$ and $dv = x \, dx$, then $du = \cos x \, dx$ and $v = x^2/2$, so integration by parts gives

$$\int x \sin x \, dx = (\sin x) \frac{x^2}{2} - \frac{1}{2} \int x^2 \cos x \, dx$$

Although this is true, $\int x^2 \cos x \, dx$ is a more difficult integral than the one we started with. In general, when deciding on a choice for u and dv , we usually try to choose $u = f(x)$ to be a function that becomes simpler when differentiated (or at least not more complicated) as long as $dv = g'(x) \, dx$ can be readily integrated to give v .

Find

$$\int x \cos x \, dx.$$

Solution We use the formula $\int u \, dv = uv - \int v \, du$ with

$$\begin{aligned} u &= x, & dv &= \cos x \, dx, \\ du &= dx, & v &= \sin x. \end{aligned}$$

Simplest antiderivative of $\cos x$

Then

$$\int x \cos x \, dx = x \sin x - \int \sin x \, dx = x \sin x + \cos x + C.$$

EXAMPLE 2 Evaluate $\int \ln x \, dx$.

SOLUTION Here we don't have much choice for u and dv . Let

$$u = \ln x \qquad dv = dx$$

Then
$$du = \frac{1}{x} dx \qquad v = x$$

Integrating by parts, we get

$$\begin{aligned} \int \ln x \, dx &= x \ln x - \int x \frac{dx}{x} \\ &= x \ln x - \int dx \\ &= x \ln x - x + C \end{aligned}$$

Integration by parts is effective in this example because the derivative of the function $f(x) = \ln x$ is simpler than f .

EXAMPLE 3 Find $\int t^2 e^t dt$.

SOLUTION Notice that t^2 becomes simpler when differentiated (whereas e^t is unchanged when differentiated or integrated), so we choose

$$u = t^2 \qquad dv = e^t dt$$

Then
$$du = 2t dt \qquad v = e^t$$

Integration by parts gives

$$\boxed{3} \qquad \int t^2 e^t dt = t^2 e^t - 2 \int t e^t dt$$

The integral that we obtained, $\int t e^t dt$, is simpler than the original integral but is still not obvious. Therefore, we use integration by parts a second time, this time with $u = t$ and

$dv = e^t dt$. Then $du = dt$, $v = e^t$, and

$$\begin{aligned}\int te^t dt &= te^t - \int e^t dt \\ &= te^t - e^t + C\end{aligned}$$

Putting this in Equation 3, we get

$$\begin{aligned}\int t^2 e^t dt &= t^2 e^t - 2 \int te^t dt \\ &= t^2 e^t - 2(te^t - e^t + C) \\ &= t^2 e^t - 2te^t + 2e^t + C_1 \quad \text{where } C_1 = -2C\end{aligned}$$

EXAMPLE 4 Evaluate $\int e^x \sin x \, dx$.

SOLUTION Neither e^x nor $\sin x$ becomes simpler when differentiated, but we try choosing $u = e^x$ and $dv = \sin x \, dx$ anyway. Then $du = e^x \, dx$ and $v = -\cos x$, so integration by parts gives

$$\boxed{4} \quad \int e^x \sin x \, dx = -e^x \cos x + \int e^x \cos x \, dx$$

The integral that we have obtained, $\int e^x \cos x \, dx$, is no simpler than the original one, but at least it's no more difficult. Having had success in the preceding example integrating by parts twice, we persevere and integrate by parts again. This time we use $u = e^x$ and $dv = \cos x \, dx$. Then $du = e^x \, dx$, $v = \sin x$, and

$$\boxed{5} \quad \int e^x \cos x \, dx = e^x \sin x - \int e^x \sin x \, dx$$

At first glance, it appears as if we have accomplished nothing because we have arrived at $\int e^x \sin x \, dx$, which is where we started. However, if we put Equation 5 into Equation 4 we get

$$\int e^x \sin x \, dx = -e^x \cos x + e^x \sin x - \int e^x \sin x \, dx$$

This can be regarded as an equation to be solved for the unknown integral. Adding $\int e^x \sin x \, dx$ to both sides, we obtain

$$2 \int e^x \sin x \, dx = -e^x \cos x + e^x \sin x$$

Dividing by 2 and adding the constant of integration, we get

$$\int e^x \sin x \, dx = \frac{1}{2}e^x(\sin x - \cos x) + C$$

Evaluate

$$\int e^x \cos x \, dx.$$

Solution Let $u = e^x$ and $dv = \cos x \, dx$. Then $du = e^x \, dx$, $v = \sin x$, and

$$\int e^x \cos x \, dx = e^x \sin x - \int e^x \sin x \, dx.$$

The second integral is like the first except that it has $\sin x$ in place of $\cos x$. To evaluate it, we use integration by parts with

$$u = e^x, \quad dv = \sin x \, dx, \quad v = -\cos x, \quad du = e^x \, dx.$$

Then

$$\begin{aligned} \int e^x \cos x \, dx &= e^x \sin x - \left(-e^x \cos x - \int (-\cos x)(e^x \, dx) \right) \\ &= e^x \sin x + e^x \cos x - \int e^x \cos x \, dx. \end{aligned}$$

The unknown integral now appears on both sides of the equation. Adding the integral to both sides and adding the constant of integration give

$$2 \int e^x \cos x \, dx = e^x \sin x + e^x \cos x + C_1.$$

Dividing by 2 and renaming the constant of integration give

$$\int e^x \cos x \, dx = \frac{e^x \sin x + e^x \cos x}{2} + C.$$

EXAMPLE Prove the reduction formula

$$\int \sin^n x \, dx = -\frac{1}{n} \cos x \sin^{n-1} x + \frac{n-1}{n} \int \sin^{n-2} x \, dx$$

where $n \geq 2$ is an integer.

SOLUTION Let $u = \sin^{n-1} x$ $dv = \sin x \, dx$

Then $du = (n-1) \sin^{n-2} x \cos x \, dx$ $v = -\cos x$

so integration by parts gives

$$\int \sin^n x \, dx = -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \cos^2 x \, dx$$

Since $\cos^2 x = 1 - \sin^2 x$, we have

$$\int \sin^n x \, dx = -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \, dx - (n-1) \int \sin^n x \, dx$$

As in Example 4, we solve this equation for the desired integral by taking the last term on the right side to the left side. Thus, we have

$$n \int \sin^n x \, dx = -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \, dx$$

or

$$\int \sin^n x \, dx = -\frac{1}{n} \cos x \sin^{n-1} x + \frac{n-1}{n} \int \sin^{n-2} x \, dx$$

Obtain a formula that expresses the integral

$$\int \cos^n x \, dx$$

in terms of an integral of a lower power of $\cos x$.

Solution We may think of $\cos^n x$ as $\cos^{n-1} x \cdot \cos x$. Then we let

$$u = \cos^{n-1} x \quad \text{and} \quad dv = \cos x \, dx,$$

so that

$$du = (n - 1) \cos^{n-2} x (-\sin x \, dx) \quad \text{and} \quad v = \sin x.$$

Integration by parts then gives

$$\begin{aligned} \int \cos^n x \, dx &= \cos^{n-1} x \sin x + (n - 1) \int \sin^2 x \cos^{n-2} x \, dx \\ &= \cos^{n-1} x \sin x + (n - 1) \int (1 - \cos^2 x) \cos^{n-2} x \, dx \\ &= \cos^{n-1} x \sin x + (n - 1) \int \cos^{n-2} x \, dx - (n - 1) \int \cos^n x \, dx. \end{aligned}$$

If we add

$$(n - 1) \int \cos^n x \, dx$$

to both sides of this equation, we obtain

$$n \int \cos^n x \, dx = \cos^{n-1} x \sin x + (n - 1) \int \cos^{n-2} x \, dx.$$

We then divide through by n , and the final result is

$$\int \cos^n x \, dx = \frac{\cos^{n-1} x \sin x}{n} + \frac{n-1}{n} \int \cos^{n-2} x \, dx.$$

$$I_n = \int \frac{dx}{(x^2+a^2)^n} = ?$$

(d)

$$I_n = \frac{1}{a^2} \int \frac{a^2 dx}{(x^2+a^2)^n} = \frac{1}{a^2} \int \frac{a^2+x^2-x^2}{(x^2+a^2)^n} dx$$

$$I_n = \frac{1}{a^2} \left\{ \int \frac{dx}{(x^2+a^2)^{n-1}} - \underbrace{\int \frac{x^2}{(x^2+a^2)^n} dx}_A \right\}$$

$$I_n = \frac{1}{a^2} \{ I_{n-1} - A \}$$

$$A = \frac{x}{2(1-n)(x^2+a^2)^{n-1}} - \frac{1}{2(1-n)} \underbrace{\int \frac{dx}{(x^2+a^2)^{n-1}}}_{I_{n-1}}$$

$$\Downarrow$$

$$\begin{matrix} x=u \\ dx=du \end{matrix}$$

$$\frac{x}{(x^2+a^2)^n} dx = dv$$

$$\frac{1}{2} \cdot \frac{1}{(-n+1)(x^2+a^2)^{n-1}} = v$$

$$I_n = \frac{1}{a^2} \left\{ I_{n-1} - \frac{x}{2(1-n)(x^2+a^2)^{n-1}} + \frac{1}{2(1-n)} I_{n-1} \right\}$$

Evaluate

$$\int x^{-3} e^{\frac{1}{x}} dx.$$

Solution: Let the integral we are trying to calculate be denoted by I . We may rewrite this as

$$I = - \int \left(\frac{-1}{x^2} \right) \left(\frac{1}{x} \right) e^{\frac{1}{x}} dx.$$

Then we make the substitution $u = \frac{1}{x}$, so $du = \frac{-1}{x^2} dx$, to yield

$$I = \int u e^u du.$$

$$I = u e^u - \int e^u du = (u - 1) e^u + C$$

where $C \in \mathbb{R}$ is the constant of integration. Substituting back to our original variable yields

$$I = \left(\frac{1}{x} - 1 \right) e^{\frac{1}{x}}$$

$$\int (\ln x)^2 \, dx$$

Using the substitution

$$u = \ln x, \quad du = \frac{1}{x} dx \quad \text{or} \quad x = e^u, \quad dx = e^u du = x du$$

the integral becomes

$$\begin{aligned} \int (\ln x)^2 \, dx &= \int u^2 e^u \, du \\ &= u^2 e^u - \int 2u^1 e^u \, du \\ &= u^2 e^u - 2u^1 e^u + \int 2 e^u \, du \\ &= (u^2 - 2u + 2) e^u + C \\ &= x ((\ln x)^2 - 2 \ln x + 2) + C \end{aligned}$$

$$\int x \ln(x+1) dx$$

Solution: We use integration by parts, taking $u = \ln(x+1)$ and $dv = x dx$. Then

$$\begin{aligned} \int x \ln(x+1) dx &= \frac{1}{2} x^2 \ln(x+1) - \frac{1}{2} \int \frac{x^2}{x+1} dx \\ &= \frac{1}{2} x^2 \ln(x+1) - \frac{1}{2} \int \left(x - 1 + \frac{1}{x+1} \right) dx \\ &= \frac{1}{2} x^2 \ln(x+1) - \frac{1}{4} x^2 + \frac{1}{2} x - \frac{1}{2} \ln(x+1) + C. \end{aligned}$$

$$\int x^2 \ln x \, dx$$

Using the Integration by Parts formula:

$$\int uv' \, dx = uv - \int u'v \, dx$$

we get:

$$\begin{aligned} \int x^2 \ln x \, dx &= (\ln x) \left(\frac{1}{3}x^3 \right) - \int \frac{1}{x} \cdot \frac{1}{3}x^3 \, dx \\ &= \frac{1}{3}x^3 \ln x - \frac{1}{3} \int x^2 \, dx \\ &= \boxed{\frac{1}{3}x^3 \ln x - \frac{1}{9}x^3 + C} \end{aligned}$$

Ex: $\int \frac{(\ln 3x)^2}{x^4} dx = ?$

$$\int \arctan x \, dx$$

We will evaluate the first integral using Integration by Parts. Let $u = \arctan x$ and $v' = 1$. Then $u' = \frac{1}{x^2 + 1}$ and $v = x$. Using the Integration by Parts formula:

$$\int uv' \, dx = uv - \int u'v \, dx$$

we get:

$$\int \arctan x \, dx = x \arctan x - \int \frac{1}{x^2 + 1} x \, dx.$$

Use the substitution $w = x^2 + 1$ to evaluate the integral on the right hand side. Then $dw = 2x \, dx \Rightarrow \frac{1}{2}dw = x \, dx$ and we get:

$$\begin{aligned} \int \arctan x \, dx &= x \arctan x - \frac{1}{2} \int \frac{1}{w} \, dw \\ &= x \arctan x - \frac{1}{2} \ln |w| + C \\ &= \boxed{x \arctan x - \frac{1}{2} \ln(x^2 + 1) + C} \end{aligned}$$

Note that the absolute value signs aren't needed because $x^2 + 1 > 0$ for all x .

$$I = \int \cos(\ln x) dx = ?$$

$$\cos(\ln x) = u \quad dx = dv$$

$$-\sin(\ln x) \cdot \frac{1}{x} dx = du \quad x = v$$

$$I = \cos(\ln x) \cdot x + \int \cancel{x} \cdot \sin(\ln x) \cdot \frac{1}{\cancel{x}} dx$$

$$\sin(\ln x) = u \quad dx = dv$$

$$\frac{1}{x} \cos(\ln x) dx = du \quad x = v$$

$$I = x \cdot \cos(\ln x) + \left\{ \sin(\ln x) \cdot x - \int \cancel{x} \cdot \frac{1}{\cancel{x}} \cos(\ln x) dx \right.$$

$$I = x \cdot \cos(\ln x) + x \cdot \sin(\ln x) - I$$

$$2I = x \cdot \cos(\ln x) + x \cdot \sin(\ln x)$$

$$I = \frac{1}{2} x \cdot \cos(\ln x) + \frac{x}{2} \cdot \sin(\ln x) + c$$

$$1) \int x \tan^{-1} x \, dx = \tan^{-1} x \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cdot \frac{dx}{1+x^2}$$

$$\begin{aligned} u &= \tan^{-1} x & dv &= x \, dx \\ du &= \frac{dx}{1+x^2} & v &= \frac{x^2}{2} \end{aligned}$$

$$= \tan^{-1} x \cdot \frac{x^2}{2} - \frac{1}{2} \int \left(1 - \frac{1}{1+x^2} \right) dx$$

$$= \tan^{-1} x \cdot \frac{x^2}{2} - \frac{1}{2} \cdot x + \frac{1}{2} \tan^{-1} x + C$$

$$\int \sin^{-1} x \, dx = \sin^{-1} x \cdot x - \int x \cdot \frac{dx}{\sqrt{1-x^2}}$$

$$\begin{aligned} u &= \sin^{-1} x & dx &= dv \\ du &= \frac{dx}{\sqrt{1-x^2}} & x &= v \end{aligned}$$

$$= x \cdot \sin^{-1} x + \frac{1}{2} \int \frac{dt}{\sqrt{t}}$$

$$= x \cdot \sin^{-1} x + \frac{1}{2} \int t^{-1/2} dt$$

$$= x \cdot \sin^{-1} x + t^{1/2} + C$$

$$= x \cdot \sin^{-1} x + \sqrt{1-x^2} + C$$

$$\begin{aligned} t &= 1-x^2 \\ dt &= -2x \, dx \\ x \, dx &= -\frac{dt}{2} \end{aligned}$$

$$\bar{I} = 2 \int (\ln x + \ln^2 x) dx = \underbrace{2 \int \ln x dx}_{I_1} + \underbrace{2 \int \ln^2 x dx}_{I_2}$$

$$I_1 : (\ln x = u \Rightarrow \frac{dx}{x} = du, dx = dv \Rightarrow x = v)$$

$$I_2 : (\ln^2 x = u \Rightarrow 2 \ln x \frac{dx}{x} = du, dx = dv \Rightarrow x = v)$$

$$\bar{I} = 2 \left\{ x \ln x - \int \cancel{x} \frac{dx}{\cancel{x}} \right\} + 2 \left\{ \ln^2 x \cdot x - \int \cancel{x} \cdot 2 \ln x \frac{dx}{\cancel{x}} \right\}$$

$$\bar{I} = 2 \{ x \cdot \ln x - x \} + 2 \{ x \cdot \ln^2 x - 2 \cdot x \ln x + 2x \}$$

$$I_n = \int x^n e^{-x} dx = -e^{-x} \cdot x^n + n \underbrace{\int e^{-x} \cdot x^{n-1} dx}_{I_{n-1}}$$

Let $u = x^n$ $dv = e^{-x} dx$
 $du = n \cdot x^{n-1} dx$ $v = -e^{-x}$

$$I_n = -x^n \cdot e^{-x} + n \cdot I_{n-1}$$

$$I_0 = \int x^0 \cdot e^{-x} dx = -e^{-x} + C$$

$$I_1 = -x e^{-x} + I_0 = -e^{-x}(x+1) + C_1$$

$$I_2 = -x^2 e^{-x} + 2I_1 = -e^{-x}(x^2 + 2x + 2) + C_2$$

$$I_3 = -x^3 e^{-x} + 3I_2 = -e^{-x}(x^3 + 3x^2 + 6x + 6) + C_3$$

⋮

$$I_n = \int x^n \cdot e^x dx = ? \quad n \geq 1$$

$$\begin{aligned} x^n &= u \\ n \cdot x^{n-1} dx &= du \end{aligned} \quad \begin{aligned} e^x dx &= dv \\ e^x &= v \end{aligned}$$

$$I_0 = \int e^x dx = e^x$$

$$I_n = x^n \cdot e^x - \underbrace{n \int x^{n-1} \cdot e^x dx}_{I_{n-1}}$$

$$\boxed{I_n = x^n \cdot e^x - n \cdot I_{n-1} + C} \quad n \geq 1$$

Örnek: $I = \int x^2 e^x dx = ?$

$$I_0 = e^x$$

$$I_2 = x^2 \cdot e^x - 2 \cdot \overline{I_1} + C$$

$$I_1 = x \cdot e^x - \overline{I_0} \rightarrow e^x$$

$$I_1 = x \cdot e^x - e^x$$

$$I_2 = x^2 \cdot e^x - 2(xe^x - e^x) + C$$

Homework 1 $I_n = \int (\ln x)^n dx = ?$